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PRACTICAL - 6

Aim: Design an interactive webpage using JavaScript for DOM manipulation, control structures, and form validation

code:

```
<!DOCTYPE html>
<html>
<head>
<title>Complaint Form</title>

<style>

body{
font-family:Arial;
background:■ #f2f2f2;
text-align:center;
}

.container{
background:■ white;
padding:20px;
width:400px;
margin:50px auto;
border-radius:10px;
box-shadow:0px 0px 10px ■ gray;
}

input,select,textarea{
width:90%;
padding:10px;
margin:10px;
}
```

```
button{
padding:10px 20px;
background:■ #28a745;
color:■ white;
border:none;
cursor:pointer;
}

.error{
color:■ red;
font-size:14px;
}

</style>

</head>

<body>

<div class="container">

<h2>Local Complaint Form</h2>

<form id="complaintForm">

<input type="text" id="name" placeholder="Enter Name">
<div id="nameError" class="error"></div>

<input type="email" id="email" placeholder="Enter Email">
<div id="emailError" class="error"></div>
```

```

<select id="issue">
  <option value="">Select Issue</option>
  <option>Garbage Problem</option>
  <option>Street Light</option>
  <option>Water Leakage</option>
  <option>Road Damage</option>
</select>
<div id="issueError" class="error"></div>

<textarea id="description" placeholder="Describe the issue"></textarea>

<button type="button" onclick="validateForm()">Submit Complaint</button>

</form>

<p id="successMsg"></p>

</div>

<script>

function validateForm(){

  let name=document.getElementById("name").value;
  let email=document.getElementById("email").value;
  let issue=document.getElementById("issue").value;

```

```

let valid=true;

document.getElementById("nameError").innerHTML="";
document.getElementById("emailError").innerHTML="";
document.getElementById("issueError").innerHTML="";
document.getElementById("successMsg").innerHTML="";

if(name===""){
  document.getElementById("nameError").innerHTML="Name is required";
  valid=false;
}

if(email===""){
  document.getElementById("emailError").innerHTML="Email is required";
  valid=false;
}

if(issue===""){
  document.getElementById("issueError").innerHTML="Please select an issue";
  valid=false;
}

if(valid){

  document.getElementById("successMsg").innerHTML="Complaint Submitted Successfully!";
  document.getElementById("successMsg").style.color="green";

}

}

</script>

```

Output:

Local Complaint Form

Garbage Problem

▼

Describe the issue

Submit Complaint

PRACTICAL-7

Aim: Build a multipage website using Bootstrap to demonstrate responsive design using the grid system and components like navbar, cards, and forms

Code:

1. index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Home</title>

  <!-- Bootstrap CDN -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>

<!-- ◆ Navbar -->
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand">MySite</a>

    <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#menu">
      <span class="navbar-toggler-icon"></span>
    </button>

    <div class="collapse navbar-collapse" id="menu">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item"><a class="nav-link active" href="index.html">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="about.html">About</a></li>
        <li class="nav-item"><a class="nav-link" href="contact.html">Contact</a></li>
```

```

        </ul>
    </div>
</div>
</nav>

<!--  Grid Section -->
<div class="container mt-5">
    <div class="row text-center">

        <div class="col-md-4 p-3 bg-primary text-white">
            Column 1
        </div>

        <div class="col-md-4 p-3 bg-success text-white">
            Column 2
        </div>

        <div class="col-md-4 p-3 bg-danger text-white">
            Column 3
        </div>

    </div>
</div>

</body>
</html>

```

2.contact.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Contact</title>

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>

```

```
<!-- Navbar -->
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand">MySite</a>

    <div>
      <a class="nav-link d-inline text-white" href="index.html">Home</a>
      <a class="nav-link d-inline text-white" href="about.html">About</a>
    </div>
  </div>
</nav>

<!-- Form -->
<div class="container mt-5">
  <h2>Contact Form</h2>

  <form>
    <div class="mb-3">
      <label>Name</label>
      <input type="text" class="form-control" placeholder="Enter name">
    </div>

    <div class="mb-3">
      <label>Email</label>
      <input type="email" class="form-control" placeholder="Enter email">
    </div>

    <div class="mb-3">
      <label>Message</label>
      <textarea class="form-control"></textarea>
    </div>

    <button class="btn btn-primary">Submit</button>
  </form>
</div>

</body>
</html>
```


3. About.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>About</title>

  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.
css" rel="stylesheet">
</head>

<body>

<!-- Navbar -->
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand">MySite</a>

    <div>
      <a class="nav-link d-inline text-white" href="index.html">Home</a>
      <a class="nav-link d-inline text-white"
href="contact.html">Contact</a>
    </div>
  </div>
</nav>

<!-- Cards -->
<div class="container mt-5">
  <div class="row">

    <div class="col-md-4">
      <div class="card">
        <div class="card-body">
          <h5 class="card-title">Card 1</h5>
          <p class="card-text">This is card content.</p>
          <button class="btn btn-primary">Read More</button>
        </div>
      </div>
    </div>
  </div>
```

```

<div class="col-md-4">
  <div class="card">
    <div class="card-body">
      <h5 class="card-title">Card 2</h5>
      <p class="card-text">This is card content.</p>
      <button class="btn btn-success">Read More</button>
    </div>
  </div>
</div>

<div class="col-md-4">
  <div class="card">
    <div class="card-body">
      <h5 class="card-title">Card 3</h5>
      <p class="card-text">This is card content.</p>
      <button class="btn btn-danger">Read More</button>
    </div>
  </div>
</div>

</div>
</div>

</body>
</html>

```

OUTPUT:

Home:



Contact:

MySite

Home About

Contact Form

Name

rohit

Email

iarhtukd@gmail.com

Message

Submit

About:

MySite

Home Contact

Card 1

This is card content.

Read More

Card 2

This is card content.

Read More

Card 3

This is card content.

Read More

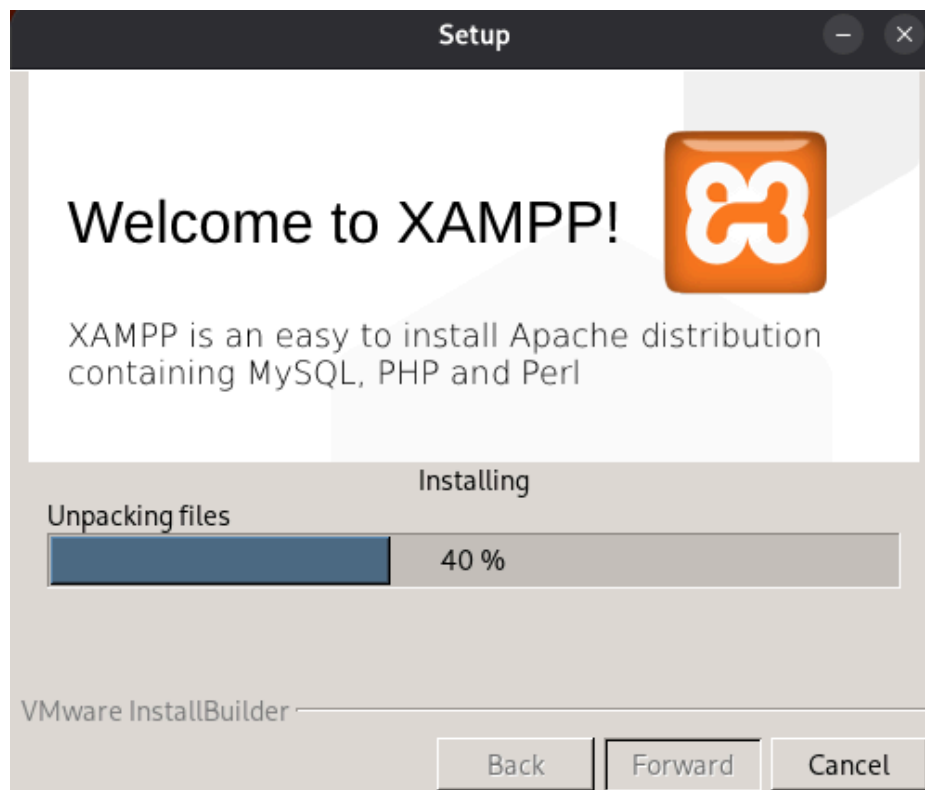
PRACTICAL - 8

Aim: Install and configure XAMPP/WAMP and write basic PHP scripts to display output, use variables, data types, and constants

Install & Configure XAMPP / WAMP

Steps:

1. Go to official site: <https://www.apachefriends.org>
2. Download XAMPP for Windows
3. Run the installer → Next → Select all components
4. Install in **C : \xampp**
5. Open **XAMPP Control Panel**
6. Start:
 - ☒ Apache
 - ☒ MySQL



TESTING THE SERVER

Open browser and type:

<http://localhost/>

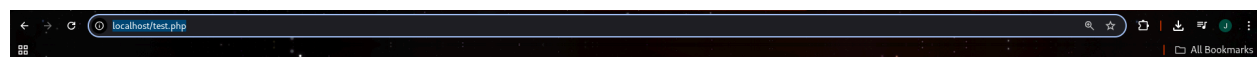


CREATING A PHP FILE

- Go to directory:
 - Linux → `/opt/lampp/htdocs/`
 - Windows → `C:\xampp\htdocs\`
- Create file: test.php
- Go to `http://localhost/test.php`

Program 1: Display Output

```
<?php
echo "Hello World!";
?>
```



Hello World!

Program 2: Using Variables

```
<?php
$name = "Jasdeep";
$age = 21;

echo "Name: " . $name;
echo "<br>";
echo "Age: " . $age;
?>
```



Name: Jasdeep
Age: 21

Program 3: Data Types

```
<?php
$intVar = 10;
$floatVar = 10.5;
$stringVar = "PHP";
$boolVar = true;

echo "Integer: " . $intVar;
echo "<br>";
echo "Float: " . $floatVar;
echo "<br>";
echo "String: " . $stringVar;
echo "<br>";
echo "Boolean: " . $boolVar;
?>
```



Integer: 10
Float: 10.5
String: PHP
Boolean: 1

Program 4: Constants

```
<?php  
define("SITE_NAME", "My Website");  
  
echo "Welcome to " . SITE_NAME;  
?>
```



Welcome to My Website

PRACTICAL - 9

Aim: Develop a PHP script to implement control structures and user-defined functions for performing logical and arithmetic operations.

Code (operations.php)

```
<?php
// ----- USER-DEFINED FUNCTIONS ----- //

// Arithmetic Functions
function add($a, $b) {
    return $a + $b;
}

function subtract($a, $b) {
    return $a - $b;
}

function multiply($a, $b) {
    return $a * $b;
}

function divide($a, $b) {
    if ($b == 0) {
        return "Error: Division by zero!";
    }
    return $a / $b;
}

// Logical Functions
function isGreater($a, $b) {
    return $a > $b;
}

function isEqual($a, $b) {
    return $a == $b;
}
```



```
// ----- INPUT VALUES ----- //
```

```
$a = 10;  
$b = 5;  
$operation = "add"; // change this to test: add, subtract, multiply,  
divide
```

```
// ----- CONTROL STRUCTURE (SWITCH) ----- //
```

```
echo "<h3>Arithmetic Operation Result:</h3>";
```

```
switch ($operation) {  
    case "add":  
        echo "Addition: " . add($a, $b);  
        break;  
  
    case "subtract":  
        echo "Subtraction: " . subtract($a, $b);  
        break;  
  
    case "multiply":  
        echo "Multiplication: " . multiply($a, $b);  
        break;  
  
    case "divide":  
        echo "Division: " . divide($a, $b);  
        break;  
  
    default:  
        echo "Invalid operation!";  
}
```

```
// ----- CONTROL STRUCTURE (IF-ELSE) ----- //
```

```
echo "<h3>Logical Operation Result:</h3>";
```

```
if (isGreater($a, $b)) {  
    echo "$a is greater than $b<br>";  
} else {  
    echo "$a is not greater than $b<br>";  
}
```

```
if (isEqual($a, $b)) {
```

```
    echo "$a is equal to $b<br>";
} else {
    echo "$a is not equal to $b<br>";
}

// ----- LOOP EXAMPLE ----- //
echo "<h3>Loop Example (Numbers from 1 to 5):</h3>";

for ($i = 1; $i <= 5; $i++) {
    echo $i . " ";
}

?>
```

OUTPUT



Arithmetic Operation Result:

Addition: 15

Logical Operation Result:

10 is greater than 5

10 is not equal to 5

Loop Example (Numbers from 1 to 5):

1 2 3 4 5

PRACTICAL - 10

Aim: Create an HTML form and use PHP to handle and validate form data using \$_GET, \$_POST, and secure input sanitization.

CODE

Form.html

```
<!DOCTYPE html>
<html>

<head>
    <title>PHP Form Example</title>
</head>

<body>

    <h2>User Registration Form</h2>

    <form method="post" action="process.php">
        Name: <input type="text" name="name"><br><br>

        Email: <input type="text" name="email"><br><br>

        Age: <input type="number" name="age"><br><br>

        <input type="submit" name="submit" value="Submit (POST)">
    </form>

    <br>

    <form method="get" action="process.php">
        Search: <input type="text" name="query">
        <input type="submit" value="Submit (GET)">
    </form>

</body>
```

```
</html>
```

Process.php

```
<?php

// Function to sanitize input
function sanitize($data) {
    $data = trim($data);           // Remove whitespace
    $data = stripslashes($data);  // Remove backslashes
    $data = htmlspecialchars($data); // Prevent XSS
    return $data;
}

// -----
// HANDLE POST REQUEST
// -----
if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $errors = [];

    // Validate Name
    if (empty($_POST["name"])) {
        $errors[] = "Name is required";
    } else {
        $name = sanitize($_POST["name"]);
        if (!preg_match("/^[a-zA-Z ]*$/", $name)) {
            $errors[] = "Only letters and spaces allowed in name";
        }
    }

    // Validate Email
    if (empty($_POST["email"])) {
        $errors[] = "Email is required";
    } else {
        $email = sanitize($_POST["email"]);
        if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
            $errors[] = "Invalid email format";
        }
    }
}
```

```

    }

}

// Validate Age
if (empty($_POST["age"])) {
    $errors[] = "Age is required";
} else {
    $age = sanitize($_POST["age"]);
    if (!filter_var($age, FILTER_VALIDATE_INT)) {
        $errors[] = "Age must be a number";
    }
}

// Output results
if (!empty($errors)) {
    echo "<h3>Errors:</h3>";
    foreach ($errors as $error) {
        echo "<p>$error</p>";
    }
} else {
    echo "<h3>Form Submitted Successfully (POST)</h3>";
    echo "Name: $name <br>";
    echo "Email: $email <br>";
    echo "Age: $age <br>";
}
}

// -----
// HANDLE GET REQUEST
// -----
if ($_SERVER["REQUEST_METHOD"] == "GET" && isset($_GET["query"])) {

    $query = sanitize($_GET["query"]);

    if (empty($query)) {
        echo "<p>Please enter a search query.</p>";
    } else {
        echo "<h3>Search Result (GET)</h3>";
        echo "You searched for: " . $query;
    }
}

```

```
}  
}  
  
?>
```

Go to <http://localhost/form.html>



User Registration Form

Name:

Email:

Age:

Search:



Form Submitted Successfully (POST)

Name: Jasdeep Singh

Email: rasna2972@gmail.com

Age: 19

PRACTICAL - 11

Aim: Design a PHP web application using include() and require() for modular coding and built-in functions for date/time and string operations.

CODE

Footer.php

```
<hr>
<footer>
    <p>&copy; <?php echo date("Y"); ?> My PHP App</p>
</footer>
</body>
</html>
```

Functions.php

```
<?php
function cleanInput($data) {
    $data = trim($data);           // remove spaces
    $data = stripslashes($data);  // remove slashes
    $data = htmlspecialchars($data); // prevent XSS
    return $data;
}

function formatName($name) {
    return ucfirst(strtolower($name)); // string functions
}

function getMessage($name) {
    date_default_timezone_set("Asia/Kolkata");
    $hour = date("H");

    if ($hour < 12) {
        return "Good Morning, $name!";
    } elseif ($hour < 18) {
```

```
        return "Good Afternoon, $name!";
    } else {
        return "Good Evening, $name!";
    }
}
?>
```

Header.php

```
<!DOCTYPE html>
<html>
<head>
    <title>PHP Modular App</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<h2>Welcome to PHP Modular Application</h2>
<hr>
```

Index.php

```
<?php require("header.php"); ?>
<?php include("functions.php"); ?>

<form method="POST" action="process.php">
    Enter Your Name: <br>
    <input type="text" name="name" required><br><br>

    Enter Your Message: <br>
    <textarea name="message" required></textarea><br><br>

    <input type="submit" value="Submit">
</form>

<?php require("footer.php"); ?>
```


Process.php

```
<?php require("header.php"); ?>
<?php require("functions.php"); ?>

<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $name = cleanInput($_POST["name"]);
    $message = cleanInput($_POST["message"]);

    $name = formatName($name);

    echo "<h3>" . getMessage($name) . "</h3>";

    echo "<p><strong>Your Message:</strong></p>";

    // String operations
    echo "<p>Uppercase: " . strtoupper($message) . "</p>";
    echo "<p>Lowercase: " . strtolower($message) . "</p>";
    echo "<p>Word Count: " . str_word_count($message) . "</p>";
    echo "<p>Reversed: " . strrev($message) . "</p>";

    // Date & Time
    echo "<p>Current Date: " . date("d-m-Y") . "</p>";
    echo "<p>Current Time: " . date("h:i:s A") . "</p>";
}
?>

<?php require("footer.php"); ?>
```

Style.css

```
body {
    font-family: Arial;
    margin: 20px;
}

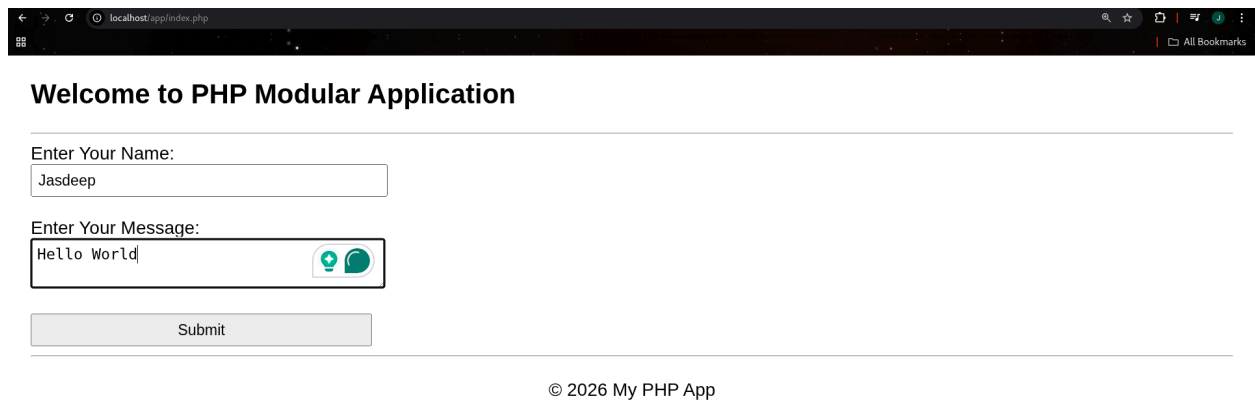
input, textarea {
```

```
width: 300px;
padding: 5px;
}

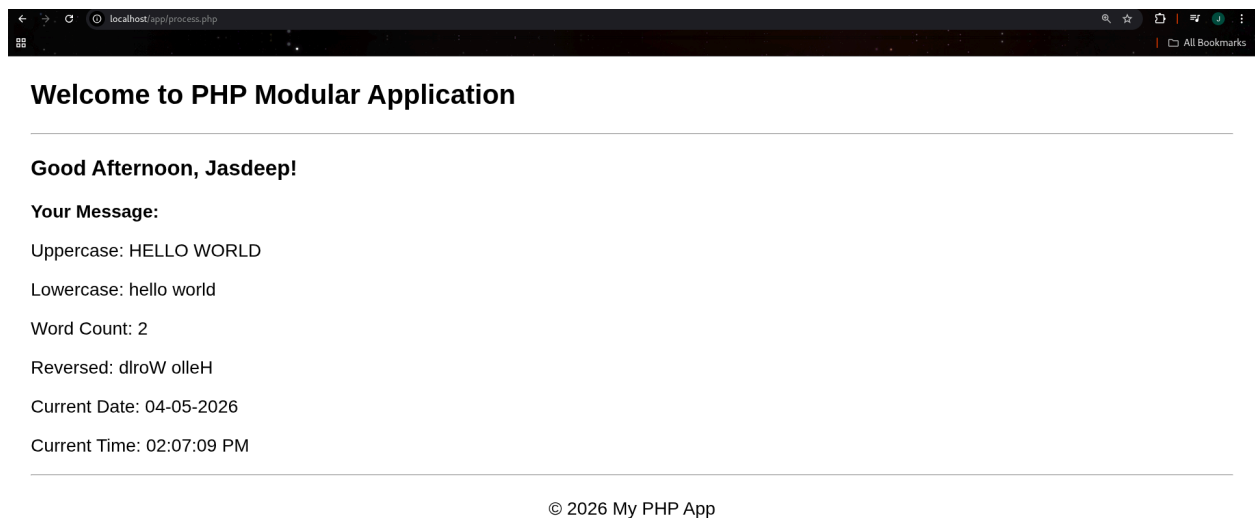
footer {
    text-align: center;
    margin-top: 20px;
}
```

OUTPUT

Go to <http://localhost/app/index.php>



A screenshot of a web browser showing the 'index.php' page of a PHP application. The page has a dark header bar with navigation links. The main content area is titled 'Welcome to PHP Modular Application' and contains a form with two input fields: 'Enter Your Name:' (with 'Jasdeep' entered) and 'Enter Your Message:' (with 'Hello World' entered). A 'Submit' button is below the message field. The footer shows '© 2026 My PHP App'.



A screenshot of a web browser showing the 'process.php' page of the same PHP application. The page is titled 'Welcome to PHP Modular Application' and displays the results of the submitted form. It shows 'Good Afternoon, Jasdeep!', 'Your Message:', and various string manipulations of the input 'Hello World' (uppercase, lowercase, word count, reversed). It also displays the current date and time. The footer shows '© 2026 My PHP App'.

PRACTICAL - 12

Aim: Build a dynamic web application using PHP and MySQL to perform CRUD operations and display data in HTML tables.

CODE

SQL Commands

```
CREATE DATABASE crud_db;
```

```
USE crud_db;
```

```
CREATE TABLE users (  
id INT AUTO_INCREMENT PRIMARY KEY,  
name VARCHAR(100),  
email VARCHAR(100)  
);
```

Db.php

```
<?php  
$conn = new mysqli("localhost", "root", "", "crud_db");  
  
if ($conn->connect_error) {  
    die("Connection failed: " . $conn->connect_error);  
}  
?>
```

Index.php

```
<?php include "db.php"; ?>  
  
<h2>User List</h2>  
<a href="add.php">Add User</a>
```

```

<table border="1">
<tr>
    <th>ID</th><th>Name</th><th>Email</th><th>Actions</th>
</tr>

<?php
$result = $conn->query("SELECT * FROM users");

while($row = $result->fetch_assoc()) {
    echo "<tr>
        <td>{$row['id']}</td>
        <td>{$row['name']}</td>
        <td>{$row['email']}</td>
        <td>
            <a href='edit.php?id={$row['id']}'>Edit</a> |
            <a href='delete.php?id={$row['id']}'>Delete</a>
        </td>
    </tr>";
}
?>
</table>

```

Add.php

```

<?php include "db.php";

if ($_POST) {
    $name = $_POST['name'];
    $email = $_POST['email'];

    $conn->query("INSERT INTO users (name, email) VALUES ('$name', '$email')");
    header("Location: index.php");
}
?>

<form method="POST">
    Name: <input type="text" name="name"><br>

```

```
Email: <input type="email" name="email"><br>
<button type="submit">Add</button>
</form>
```

Edit.php

```
<?php include "db.php";

$id = $_GET['id'];
$result = $conn->query("SELECT * FROM users WHERE id=$id");
$row = $result->fetch_assoc();

if ($_POST) {
    $name = $_POST['name'];
    $email = $_POST['email'];

    $conn->query("UPDATE users SET name='$name', email='$email' WHERE
id=$id");
    header("Location: index.php");
}
?>

<form method="POST">
    Name: <input type="text" name="name" value="<?= $row['name'] ?>"><br>
    Email: <input type="email" name="email" value="<?= $row['email']
?>"><br>
    <button type="submit">Update</button>
</form>
```

Delete.php

```
<?php include "db.php";

$id = $_GET['id'];
$conn->query("DELETE FROM users WHERE id=$id");
```

```
header("Location: index.php");  
?>
```

OUTPUT



User List

[Add User](#)

ID	Name	Email	Actions
----	------	-------	---------



Name:

Email:



User List

[Add User](#)

ID	Name	Email	Actions
1	Jasdeep Singh	rasna2972@gmail.com	Edit Delete



Name:

Email: